

Chapter 08

Spectral Engineering and Thermal Windows

Controlling Heat by Controlling What Photons Can Do

Thermal Photonics: A Unified Framework for Engineering Heat Flux
Book 3 of the Open-Source Engineering Optics Trilogy

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Learning Objectives

- LO-08.1** Distinguish intrinsic, structured, and effective emissivity, and explain why emissivity replaces conductivity as the primary design variable for radiation-dominated paths ($\Gamma > 10$).
- LO-08.2** Design thin-film interference stacks for spectral shaping of thermal emission using quarter-wave theory and high-index IR materials (Ge, ZnSe, PbTe, chalcogenides).
- LO-08.3** Select appropriate spectral profiles from the design catalog for four canonical semiconductor applications: RTP heating, pyrometry windows, radiative cooling, and TPV emitters.
- LO-08.4** Design cold mirrors and dichroic filters that decouple radiative heat from visible or diagnostic wavelengths in process chambers.
- LO-08.5** Evaluate high-temperature coating stability and select materials for operation above 600 °C using the thermal stability figure of merit.
- LO-08.6** Calculate thermal lensing in IR windows using the dn/dT -based focal length formula and the thermal lensing figure of merit $F_{TL} = \kappa/|dn/dT|$.
- LO-08.7** Integrate spectral engineering elements into the Mode-Path Matrix as radiation conductance elements $G_{\text{rad}} = 4\varepsilon_{\text{eff}}\sigma_{\text{SB}}T^3A$, including parallel conduction–radiation paths through windows.
- LO-08.8** Perform sensitivity analysis of G_{rad} with respect to emissivity and temperature, and apply the recomputation threshold (DR-10.6) to determine when spectral re-evaluation is required.

Abbreviations and Nomenclature

Abbreviation	Meaning
AR	Anti-reflection
BSR	Back-surface reflector
CVD	Chemical vapor deposition
DOE	Diffraction optical element
FTIR	Fourier-transform infrared
FOM	Figure of merit

Abbreviation	Meaning
GASIR	Ge–As–Se IR glass (commercial)
LWIR	Long-wave infrared (8–14 μm)
MPM	Mode-Path Matrix
MWIR	Mid-wave infrared (3–5 μm)
OPD	Optical path difference
PGM	Precision glass molding
PVD	Physical vapor deposition
QW	Quarter-wave
RMS	Root mean square
RTP	Rapid thermal processing
SWIR	Short-wave infrared (1–3 μm)
TPV	Thermophotovoltaic
WFE	Wavefront error

Symbol	Meaning	SI Unit
$\varepsilon(\lambda)$	Spectral emissivity	—
ε_{eff}	Effective (band-averaged) emissivity	—
$\alpha(\lambda)$	Spectral absorptivity	—
$\rho(\lambda)$	Spectral reflectance	—
$\tau(\lambda)$	Spectral transmittance	—
n	Refractive index	—
dn/dT	Thermo-optic coefficient	K^{-1}
κ	Thermal conductivity	$\text{W m}^{-1} \text{K}^{-1}$
F_{TL}	Thermal lensing FOM = $\kappa/ dn/dT $	W m^{-1}
f_{th}	Thermal lens focal length	m
Γ	Mode dominance ratio = $G_{\text{rad}}/G_{\text{cond}}$	—
G_{rad}	Radiation conductance	W K^{-1}
G_{cond}	Conduction conductance	W K^{-1}
σ_{SB}	Stefan–Boltzmann constant	$\text{W m}^{-2} \text{K}^{-4}$
$B(\lambda, T)$	Planck spectral radiance	$\text{W m}^{-2} \text{sr}^{-1} \text{m}^{-1}$
F_{12}	View factor	—
T_{max}	Maximum operating temperature	K
w	Beam radius (at window)	m
P	Absorbed optical power	W
S	Strehl ratio	—

Subscript/Superscript	Convention
λ	Spectral (wavelength-dependent)
eff	Effective (band-averaged or system-level)
struct	Structured (by thin-film or metasurface)
intr	Intrinsic (bulk material property)
rad	Radiative
cond	Conductive
sol	Solar
atm	Atmospheric (8–13 μm window)

Numbering convention for this chapter: Design rules use the **DR-10.n** prefix (inherited from v2.2 Chapter 10 for continuity; see the note in §08.1 for justification). Worked examples use **WE-08.n**. Learning objectives use **LO-08.n**.

Chapter Roadmap and Deliverables

This chapter is organized around a single engineering deliverable:

A complete spectral engineering methodology—from emissivity specification through material selection and thin-film stack design to thermal window co-design—integrated into the Mode-Path Matrix framework, with the linearized radiation conductance $G_{\text{rad}} = 4 \varepsilon_{\text{eff}} \sigma_{\text{SB}} T^3 A$ as the quantitative output for every spectral element.

To help you navigate, Figure 08.1 shows the chapter “spine” (workflow) and how each section feeds into the final spectral engineering toolkit.

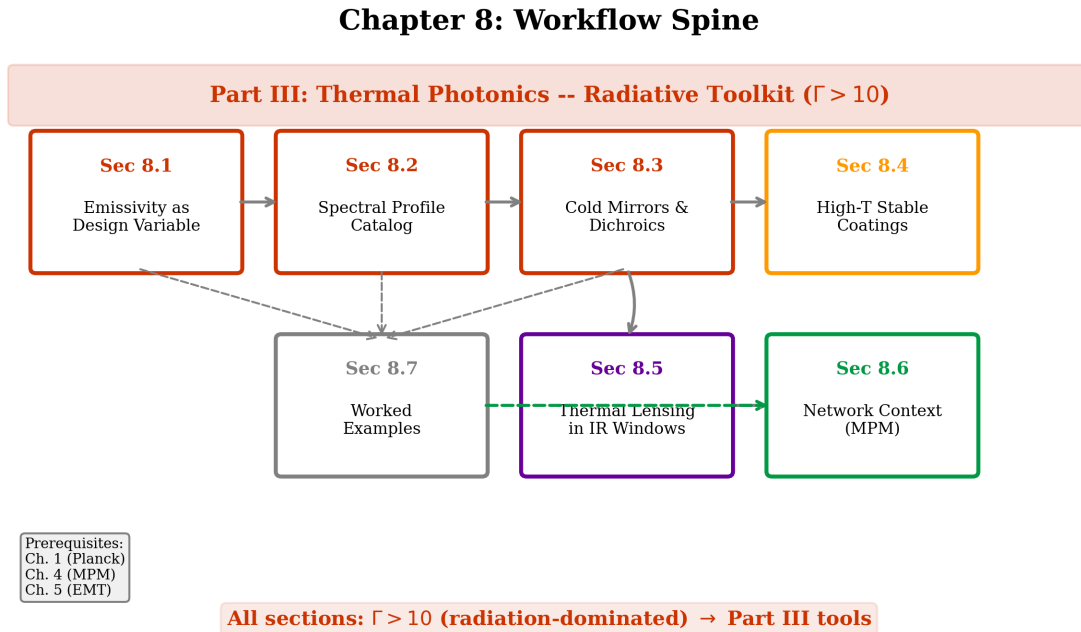


Figure 08.1: Chapter 08 roadmap. The narrative spine begins with emissivity as the primary radiation design variable (§08.1), builds a spectral profile catalog for semiconductor applications (§08.2), then develops the dichroic and cold-mirror tools for decoupling heat from diagnostics (§08.3). Metalenses and DOEs (§08.4) add spatial control. High-temperature material stability (§08.5) and thermal lensing (§08.6) address the two dominant failure modes for IR windows in production environments. Section 08.7 integrates all spectral elements into the Mode-Path Matrix as G_{rad} network elements. Red boxes indicate radiation-dominated content ($\Gamma > 10$); the green network context box connects to the architectural framework of Chapter 4.

Part III: Thermal Photonics—The Radiative Toolkit

Part II (Chapters 5–7) equipped the reader with the Diffusionics toolkit for conduction-dominated paths ($\Gamma < 0.1$): transformation thermotics shaped the conductivity tensor, effective medium theory replaced microstructure with bulk anisotropy, and via arrays delivered production-ready metamaterials. Every tool in Part II modified the conduction conductance $G_{\text{cond}} = \kappa A/L$ in the thermal network.

Part III now turns to the *complementary* toolkit. When the Mode-Path Matrix identifies a radiation-dominated path ($\Gamma > 10$), the engineer's primary design variable shifts from the conductivity tensor κ to the spectral emissivity $\varepsilon(\lambda)$. The governing equation changes from Fourier's law to the Planck distribution, and the design tools shift from composite materials to thin-film interference, selective emitters, and spectral filters. The radiative conductance $G_{\text{rad}} = 4 \varepsilon_{\text{eff}} \sigma_{\text{SB}} T^3 A$ is the network element that Part III modifies.

This chapter opens Part III by establishing the spectral engineering fundamentals. Chapter 9 applies them to radiative cooling, shielding, and the mixed-mode regime. Chapter 10 extends to thermophotovoltaics and energy recovery.

Reader Recipe

MPM-first workflow for this chapter:

1. Confirm that the path of interest is radiation-dominated: compute $\Gamma > 10$ from Chapter 4.
2. Identify the target spectral band from the Wien peak $\lambda_{\text{max}} = b/T$ (Chapter 1, Eq. 01.9) and the application-specific spectral profile (§08.2, Table 08.6).
3. Select the coating architecture (§08.3–§08.4) and compute the resulting ε_{eff} .
4. Calculate the modified G_{rad} and update the Mode-Path Matrix (§08.7).
5. If the window introduces a parallel conduction path, add it; if thermal lensing exceeds tolerance, apply §08.6 corrections.

Pain Point

Why this chapter matters:

An engineer who masters Part II (conduction shaping) but neglects Part III (radiation control) will encounter five production failure modes at temperatures above 600 °C:

- F1. Radiative parasitic undermining uniformity** (§08.7, WE-08.3): Uncontrolled emissivity on chamber walls creates parasitic G_{rad} paths that degrade wafer temperature uniformity by > 10 K during RTP annealing, even when the conduction network is perfectly optimized.
- F2. Thermal lensing corrupting overlay** (§08.6, WE-08.1): Every IR window absorbs power and develops a thermal lens. A coating defect that increases α by $100\times$ drives the wavefront error from $\lambda/390$ to $\lambda/4$ —a catastrophic failure for overlay alignment.
- F3. Coating degradation at high temperature** (§08.5): Ge coatings oxidize above 200 °C ($n : 4.0 \rightarrow 1.6$); PbTe/ZnS stacks interdiffuse above 250 °C; chalcogenides crystallize near T_g . Ignoring thermal limits destroys spectral selectivity within hours.
- F4. Pyrometry errors from uncharacterized ε** (§08.2): Pyrometer readings scale as $T \propto [1/\varepsilon(\lambda_s)]^{1/4}$; a 20% error in ε at the sensor wavelength produces a $\sim 5\%$ error in inferred T at 1000 K—approximately 50 K.
- F5. Spectral profile mismatch** (§08.2): Using a broadband emitter where a selective profile is required wastes $> 60\%$ of radiated power in parasitic sub-bandgap emission (WE-08.2 demonstrates this for TPV applications).

The Mode-Path Matrix identifies *which* paths are radiation-dominated; this chapter provides the engineering tools to *control* those paths.

Part II concluded with via array architectures that give the engineer complete control over where heat flows by conduction. But for processes above approximately 600 °C—RTP annealing, epitaxial growth, thermal oxidation, and laser processing—conduction control is necessary but insufficient. At these temperatures, radiative heat exchange carries a comparable or dominant

fraction of the thermal budget, and the Mode-Path Matrix classifies these paths as $\Gamma > 10$. Controlling heat in this regime requires controlling photons: which wavelengths are emitted, which are absorbed, which are reflected, and which are transmitted.

The engineer who masters Part II but ignores Part III will design a perfectly optimized conduction network—and then watch radiative parasitic paths undermine the thermal uniformity by tens of degrees during a 1000 °C anneal. Conversely, a radiation shield that blocks thermal emission but inadvertently creates an uncompensated thermal lensing element in the process chamber optics introduces wavefront errors that corrupt overlay alignment.

This chapter unifies spectral emissivity engineering with thermal window design, recognizing that in semiconductor fabrication, every optical element in a high-temperature chamber is simultaneously a thermal element. The Mode-Path Matrix demands that we treat it as such.

08.1 Emissivity as a Design Variable

A note on design rule numbering: The design rules in this chapter carry the prefix DR-10.n, inherited from v2.2 Chapter 10 (“Spectral Engineering Fundamentals”). The v3.0 consolidation merged v2.2 Chapters 10 and 11 into this single Chapter 8, but the DR-10 prefix is retained to maintain backward compatibility with cross-references in the companion quick-reference card (`tpd-mpm-quick-ref`) and the Problem Solutions manual. When using the MPM quick-reference card alongside this chapter, the DR-10.n labels match directly.

08.1.1 The Transition from κ to $\varepsilon(\lambda)$

Throughout Part II, the central quantity was the thermal conductivity tensor κ : binary layers, transformation maps, and via arrays all aimed to sculpt κ into a desired spatial profile. In Part III, the central quantity becomes the spectral emissivity $\varepsilon(\lambda, \theta)$.

Kirchhoff’s law (Chapter 1, Eq. 01.12) guarantees that at thermal equilibrium, $\varepsilon(\lambda, \theta) = \alpha(\lambda, \theta)$: a surface engineered to emit strongly at a target wavelength also absorbs strongly there, and one engineered to be a poor emitter is necessarily a good reflector. Energy conservation for an opaque surface further constrains:

$$\varepsilon(\lambda) + \rho(\lambda) = 1 \quad (08.1)$$

where ρ is the spectral reflectance. For a semi-transparent window or substrate, the full constraint becomes:

$$\varepsilon(\lambda) + \rho(\lambda) + \tau(\lambda) = 1 \quad (08.2)$$

where τ is the spectral transmittance.

Radiation Mode

The fundamental design duality of Part III:

Increasing ε at a target wavelength increases *both* emission and absorption. This is simultaneously a tool and a constraint:

- **Tool:** High ε in the atmospheric window (8–13 μm) enables radiative cooling.
- **Constraint:** The same high ε absorbs ambient thermal radiation, reducing net cooling power.

Spectral engineering resolves this tension by making ε wavelength-selective—high where emission is desired, low where absorption is parasitic.

08.1.2 Intrinsic, Structured, and Effective Emissivity

The total emissivity of a thermal surface has three contributions that the engineer can manipulate independently:

1. **Intrinsic emissivity** $\varepsilon_{\text{intr}}(\lambda)$: determined by the bulk optical constants (n, k) of the surface material via the Fresnel equations. For a smooth semi-infinite slab at normal incidence:

$$\varepsilon_{\text{intr}}(\lambda) = 1 - \left| \frac{n(\lambda) + ik(\lambda) - 1}{n(\lambda) + ik(\lambda) + 1} \right|^2 \quad (08.3)$$

Metals ($n \sim 1$, $k \gg 1$) have $\varepsilon_{\text{intr}} < 0.1$; dielectrics ($k \approx 0$) have $\varepsilon_{\text{intr}}$ determined by n alone, with higher n giving lower emissivity. At $n = 4$ (Ge), $\varepsilon_{\text{intr}} = 0.36$ at normal incidence.

2. **Structured emissivity** $\varepsilon_{\text{struct}}(\lambda)$: created by thin-film interference, photonic crystals, metasurfaces, or surface texturing. These add wavelength-dependent phase accumulation that modifies $\rho(\lambda)$ and hence $\varepsilon(\lambda)$ via Eq. (08.1).

3. **Effective emissivity** ε_{eff} : the Planck-weighted band average that enters the linearized radiation conductance:

$$\varepsilon_{\text{eff}}(T) = \frac{\int_0^\infty \varepsilon(\lambda) B(\lambda, T) d\lambda}{\int_0^\infty B(\lambda, T) d\lambda} \quad (08.4)$$

This is the quantity that appears in the MPM network element $G_{\text{rad}} = 4 \varepsilon_{\text{eff}} \sigma_{\text{SB}} T^3 A$.

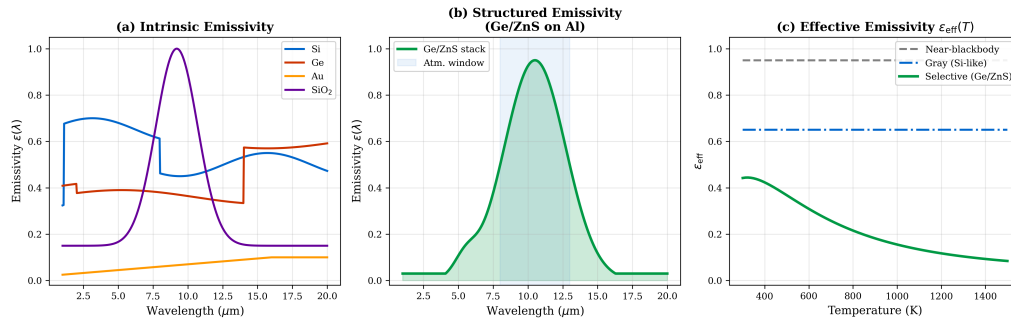


Figure 08.2: Emissivity engineering taxonomy. **Left:** Intrinsic emissivity of representative materials (polished Si, Ge, Au, SiO₂) from 1 to 20 μm , governed by bulk optical constants. **Center:** Structured emissivity created by a Ge/ZnS multilayer stack on an Al substrate—the thin-film interference sculpts a high- ε band within the 8–13 μm atmospheric window while maintaining low ε outside it. **Right:** Effective emissivity $\varepsilon_{\text{eff}}(T)$ computed from Eq. (08.4) for three surface designs, showing how spectral selectivity improves radiative performance at specific temperatures.

Design Rule: DR-10.1 — Emissivity as Network Element

For every radiation-dominated path ($\Gamma > 10$) in the Mode-Path Matrix, the design variable is the effective emissivity $\varepsilon_{\text{eff}}(T)$, not the conductivity κ . Compute ε_{eff} from Eq. (08.4) using the spectral emissivity profile and the Planck function at the operating temperature. The resulting network conductance is $G_{\text{rad}} = 4 \varepsilon_{\text{eff}} \sigma_{\text{SB}} T^3 A$. For broadband surfaces ($\varepsilon \approx \text{const.}$), $\varepsilon_{\text{eff}} \approx \varepsilon$; for spectrally selective surfaces, ε_{eff} is temperature-dependent and must be recalculated if T changes by more than 50 K.

08.1.3 Sensitivity of G_{rad} to ε_{eff} and T

The linearized radiation conductance $G_{\text{rad}} = 4 \varepsilon_{\text{eff}} \sigma_{\text{SB}} T^3 A$ depends on two independently controllable quantities: the effective emissivity ε_{eff} (set by coating design) and the operating tem-

perature T (set by the process). Understanding their relative sensitivity is essential for deciding where to invest engineering effort.

Emissivity sensitivity. At fixed T and A :

$$\frac{\partial G_{\text{rad}}}{\partial \varepsilon_{\text{eff}}} = 4 \sigma_{\text{SB}} T^3 A \quad (08.5)$$

This is *linear* in ε_{eff} : changing ε_{eff} from 0.5 to 0.1 reduces G_{rad} by exactly 80%, independent of temperature. Emissivity is the precision knob.

Temperature sensitivity. At fixed ε_{eff} and A :

$$\frac{\partial G_{\text{rad}}}{\partial T} = 12 \varepsilon_{\text{eff}} \sigma_{\text{SB}} T^2 A \quad (08.6)$$

The T^2 dependence means that temperature fluctuations become increasingly dangerous at high T . The relative sensitivity ratio is:

$$\frac{\partial G_{\text{rad}}/G_{\text{rad}}}{\partial T/T} = 3 \quad (08.7)$$

A 1% increase in temperature produces a 3% increase in G_{rad} —three times more sensitive than a 1% change in ε_{eff} .

Numerical example (at $T = 1100$ K, $\varepsilon_{\text{eff}} = 0.7$, $A = 0.07$ m²):

$$\begin{aligned} G_{\text{rad}} &= 4 \times 0.7 \times 5.67 \times 10^{-8} \times 1100^3 \times 0.07 = 4 \times 0.7 \times 5.67 \times 10^{-8} \times 1.331 \times 10^9 \times 0.07 \\ &= 4 \times 0.7 \times 5.67 \times 10^{-8} \times 9.317 \times 10^7 = 4 \times 0.7 \times 5.283 \\ &= 14.79 \text{ W/K} \end{aligned} \quad (08.8)$$

$$\frac{\partial G_{\text{rad}}}{\partial \varepsilon_{\text{eff}}} = \frac{14.79}{0.7} = 21.1 \text{ W/K per unit } \varepsilon_{\text{eff}} \quad (08.9)$$

$$\frac{\partial G_{\text{rad}}}{\partial T} = \frac{3 \times 14.79}{1100} = 0.0403 \text{ W/K per K} \quad (08.10)$$

Interpretation: changing ε_{eff} by 0.1 (a one-layer coating modification) changes G_{rad} by 2.1 W/K; a 50 K temperature drift changes G_{rad} by 2.0 W/K—comparable magnitudes. This means that at high temperatures, both coating specification *and* temperature control are first-order concerns.

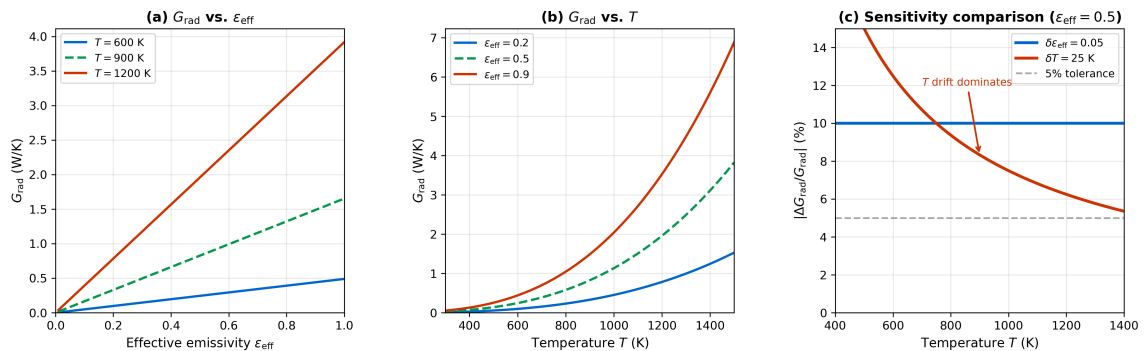


Figure 08.3: Sensitivity of G_{rad} to emissivity and temperature. **(a)** G_{rad} vs. ε_{eff} at three temperatures ($T = 600, 900, 1100$ K), showing the linear scaling with emissivity and the dramatic increase with temperature. $A = 0.07$ m². **(b)** $\partial G_{\text{rad}}/\partial \varepsilon_{\text{eff}}$ vs. T , demonstrating the T^3 dependence of emissivity sensitivity. **(c)** $\partial G_{\text{rad}}/\partial T$ vs. T at $\varepsilon_{\text{eff}} = 0.3, 0.5, 0.7$, showing the T^2 growth that makes temperature drift increasingly dangerous at high process temperatures. The 50 K recomputation threshold (DR-10.6) is marked.

Design Rule: DR-10.6 — Sensitivity Recomputation Threshold

For spectrally selective surfaces, recompute $\varepsilon_{\text{eff}}(T)$ and update G_{rad} in the MPM whenever:

- Temperature changes by more than 50 K from the design point, **or**
- Emissivity drifts by more than $|\Delta\varepsilon_{\text{eff}}| > 0.05$ due to coating degradation, contamination, or aging.

If either threshold is exceeded, the network conductances may have shifted by $> 10\%$, potentially changing the mode classification (Γ crossing 0.1 or 10) and invalidating the tool selection.

08.1.4 Thin-Film Interference for Spectral Shaping

The most production-mature method for engineering $\varepsilon_{\text{struct}}(\lambda)$ is multilayer thin-film interference. A stack of alternating high-index (n_H) and low-index (n_L) dielectric layers creates constructive and destructive interference that shapes the reflectance spectrum $\rho(\lambda)$, and hence $\varepsilon(\lambda) = 1 - \rho(\lambda)$ for opaque substrates.

The fundamental building block is the **quarter-wave layer**: a film of optical thickness $n \cdot d = \lambda_0/4$, where λ_0 is the design wavelength. At λ_0 , reflections from the front and back surfaces interfere constructively (for the reflected beam), creating a high-reflectance band centered at λ_0 .

Trilogy Connection: Eikonal Bridge Ch. 3, 8

The thin-film interference physics here is mathematically identical to the coating design treated in Book 1 (*The Eikonal Bridge*, Chapters 3 and 8). The transfer-matrix method, Herpin equivalent layers, and admittance diagram tools developed there apply directly. The only change is the operating wavelength: visible/NIR in Book 1, MWIR/LWIR in this chapter.

The **fractional bandwidth** of a quarter-wave stack reflector is governed by the index contrast:

$$\frac{\Delta\lambda}{\lambda_0} = \frac{4}{\pi} \arcsin\left(\frac{n_H - n_L}{n_H + n_L}\right) \quad (08.11)$$

High-index contrast pairs (Ge/ZnS: $n_H/n_L = 4.0/2.2 = 1.82$; PbTe/ZnS: $5.5/2.2 = 2.50$) achieve 30–43% fractional bandwidth per stopband, versus $< 20\%$ for typical visible-band coatings. This wider bandwidth is a direct consequence of the high refractive indices available in the IR (Table 08.4).

Table 08.4: Material pairs for IR thin-film spectral engineering. Bandwidth from Eq. (08.11). Index values at $\lambda = 10 \mu\text{m}$.

n_H	Material	n_L	Material	n_H	n_L	n_H/n_L	$\Delta\lambda/\lambda_0$
PbTe		ZnS		5.5	2.2	2.50	0.426
Ge		ZnS		4.0	2.2	1.82	0.315
Ge		CaF ₂		4.0	1.3	3.08	0.496
ZnSe		BaF ₂		2.4	1.4	1.71	0.289
Si		SiO ₂		3.4	1.4	2.43	0.416

The required physical thickness of a quarter-wave layer scales inversely with refractive index:

$$d_{\text{QW}} = \frac{\lambda_0}{4n} \quad (08.12)$$

At $\lambda_0 = 10 \mu\text{m}$: $d_{\text{QW}}(\text{PbTe}) = 0.45 \mu\text{m}$, $d_{\text{QW}}(\text{CaF}_2) = 1.92 \mu\text{m}$. High-index materials enable thinner layers, reducing total stack thickness, deposition time, and interfacial stress—critical for production reliability.

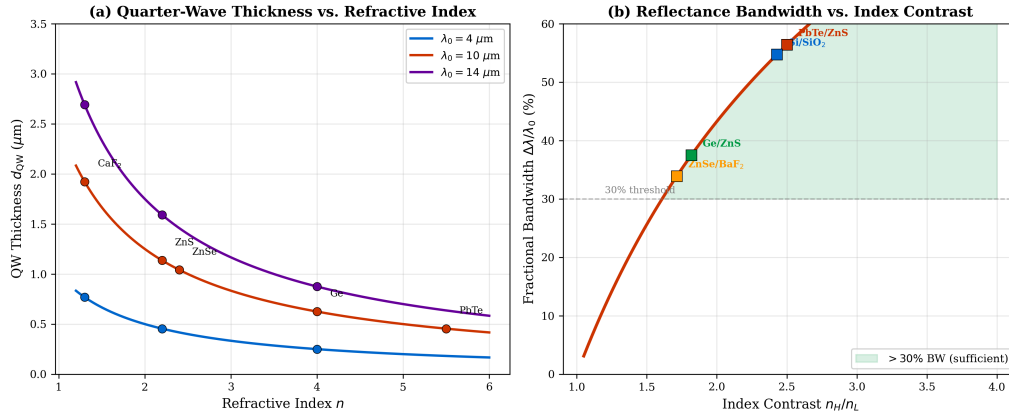


Figure 08.4: Thin-film design space for IR spectral engineering. **Left:** Quarter-wave thickness vs. refractive index for three design wavelengths ($\lambda_0 = 4, 10, 14 \mu\text{m}$). High- n materials (PbTe, Ge) enable sub-micron layers even at LWIR wavelengths. **Right:** Reflectance bandwidth $\Delta\lambda/\lambda_0$ vs. index contrast n_H/n_L from Eq. (08.11). The shaded region marks pairs achieving $> 30\%$ fractional bandwidth, sufficient for most single-band thermal applications.

Why high-index materials dominate IR spectral engineering. The combination of Eqs. (08.11)–(08.12) creates a compound advantage for high-index materials in the MWIR and LWIR:

1. Thinner layers $\rightarrow$ lower stress, faster deposition, better thickness control.
2. Wider stopbands $\rightarrow$ fewer layers needed for a given rejection ratio.
3. Higher phase coverage $\rightarrow$ enables metasurface elements with sub-wavelength pitch.

Box 8.1 quantifies this advantage for metasurface design.

08.1.5 Coating Production Specification

Spectral coating designs that perform well in simulation must survive the production environment. Every coating specified for a thermal photonics application must pass the seven-parameter acceptance test in Table 08.5.

Table 08.5: Seven-parameter coating production specification. Every spectral coating must satisfy all seven criteria before deployment in a thermal photonics application.

#	Parameter	Requirement	Test Method
1	ε_{eff} at T_{design}	Within ± 0.03 of target	FTIR + Planck integration
2	Spectral bandwidth	$\Delta\lambda/\lambda_0$ meets profile	FTIR reflectance
3	Angular tolerance	Spec met to θ_{max}	Variable-angle FTIR
4	Thermal cycling	$ \Delta\varepsilon_{\text{eff}} < 0.02$ after 10^3 cycles to T_{max}	Furnace cycle + FTIR
5	Film stress	$\sigma < \sigma_{\text{crit}}$ (Ch. 5 DR-05.7)	Wafer bow measurement
6	Adhesion	Pass tape test (ASTM D3359-B, 5B rating) after thermal cycle	Tape pull
7	Emissivity drift	$ \Delta\varepsilon_{\text{eff}} < 0.02$ after 10^4 hours at $0.8 T_{\text{max}}$	Accelerated aging

Design Rule: DR-10.7 — Coating Production Acceptance

Every spectral coating deployed in a thermal photonics application must pass the seven-parameter specification in Table 08.5, including thermal cycling ($|\Delta\varepsilon_{\text{eff}}| < 0.02$ after 10^3 cycles), film stress below the critical cracking threshold (Chapter 5, DR-05.7), and long-term emissivity drift. Coatings that pass optical specification but fail mechanical or thermal-cycling criteria are *not* production-ready.

Fabrication Challenges**Three hidden emissivity drift mechanisms:**

1. **Native oxide growth:** Even “stable” surfaces develop thin native oxides over time. On Ge, a 2 nm GeO_2 layer shifts the effective index from $n = 4.0$ to an effective $n \approx 3.6$ at the surface, detuning quarter-wave stacks by $\sim 3\%$ in center wavelength.
2. **Surface roughness evolution:** Thermal cycling induces grain growth in polycrystalline films, increasing RMS roughness. When $\sigma_{\text{RMS}} > \lambda/20$, scattering losses reduce the effective reflectance and increase the apparent emissivity.
3. **Cavity effects in multilayer stacks:** Delamination or void formation at interfaces creates parasitic Fabry–Pérot cavities that introduce spurious spectral features, corrupting the designed $\varepsilon(\lambda)$ profile. These are detectable by FTIR monitoring but invisible to broadband pyrometry.

All three mechanisms are captured by the emissivity drift criterion (parameter 7 in Table 08.5).

08.2 Spectral Profile Design Catalog

The spectral engineering tools of §08.1 are general-purpose. This section applies them to four canonical semiconductor thermal applications, each requiring a distinct spectral emissivity profile. Table 08.6 is the central reference for Part III spectral design.

Table 08.6: Spectral profile design catalog for semiconductor thermal applications. Each row specifies the target emissivity profile, material recommendation, and the resulting network modification.

Application	Required file	Pro-	Material System	MPM Network Effect
RTP heating	High ε at 0.9–1.1 μm		Si/SiO ₂ multilayer	Maximizes G_{rad} at lamp band; reduces ε_{eff} outside
Pyrometry window	Low ε at sensor λ ; high τ		Ge/ZnS quarter-wave stack	Creates transparent spectral channel; $\tau \rightarrow 1$ at sensor band
Radiative cooling	High ε at 8–13 μm ; low ε at solar		As ₂ S ₃ metasurface + Al	Maximizes G_{rad} in atmospheric window; minimizes solar absorption
TPV emitter	Narrowband ε at E_g		W/HfO ₂ cavity	Concentrates G_{rad} at PV bandgap; suppresses sub-bandgap loss

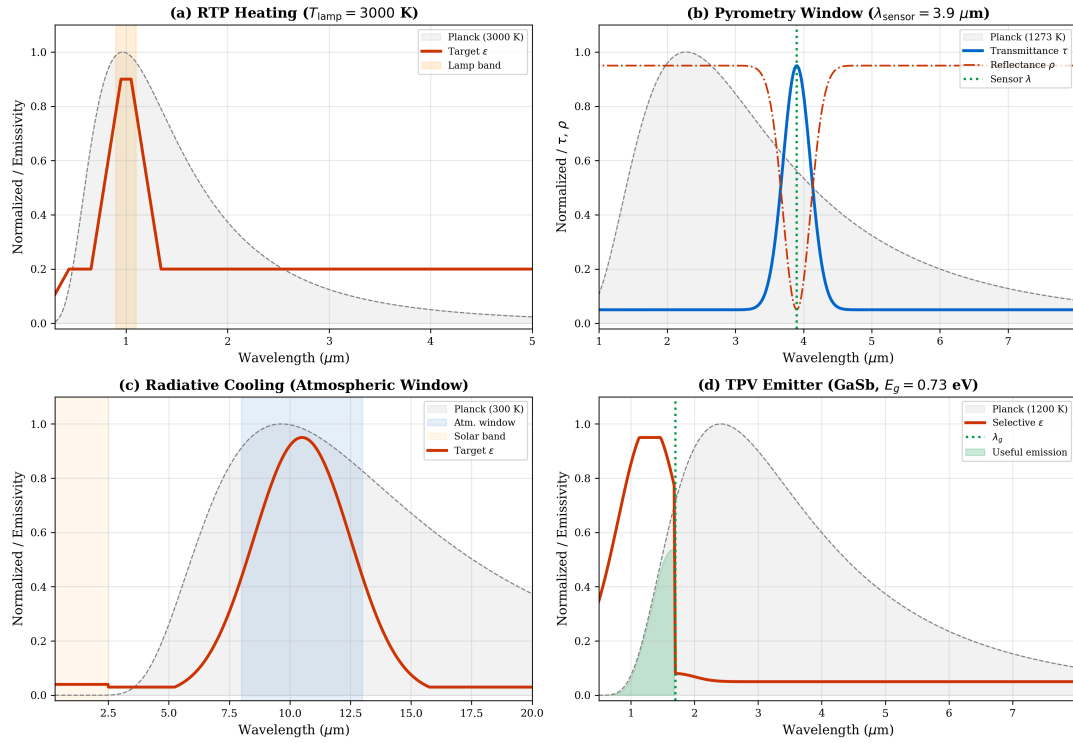


Figure 08.5: Spectral profile catalog for the four canonical semiconductor applications in Table 08.6. Each panel shows the target emissivity profile (red), the Planck spectrum at the operating temperature (gray fill), and the atmospheric transmission window (blue shading for 8–13 μm). The overlap between $\varepsilon(\lambda)$ and $B(\lambda, T)$ determines ε_{eff} via Eq. (08.4).

08.2.1 RTP Heating: Maximizing Lamp Coupling

Rapid thermal processing uses tungsten-halogen lamps with peak emission near $\lambda = 0.9\text{--}1.1\ \mu\text{m}$ (Wien peak at $T_{\text{lamp}} \approx 3000\ \text{K}$). The wafer must absorb efficiently in this band while the pyrometer sensor (typically at $\lambda = 0.95\ \mu\text{m}$ or $3.3\ \mu\text{m}$) must see the wafer emissivity, not lamp reflections.

The challenge: bare silicon is partially transparent below the bandgap ($\lambda < 1.1\ \mu\text{m}$), and its emissivity depends on doping, temperature, and surface condition. A Si/SiO₂ multilayer stack on the backside can be designed to enhance absorption at the lamp band while suppressing stray reflections.

Design Rule: DR-10.2 — Spectral Profile Selection

Match the spectral emissivity profile to the application using the design catalog (Table 08.6). For every radiation path in the MPM where $\Gamma > 10$, select the profile that maximizes the desired G_{rad} component (absorption, emission, or transmission) while minimizing parasitic components. Verify that the resulting $\varepsilon_{\text{eff}}(T)$ remains within the linearization validity range (DR-T.1b: $\Delta T/\bar{T} < 0.2$).

08.2.2 Pyrometry Windows: Creating Diagnostic Channels

In-situ pyrometry requires a spectral window through which the sensor receives thermal radiation from the wafer surface without contamination by lamp radiation, chamber wall emission, or window self-emission. The ideal pyrometry window has:

- High transmittance ($\tau > 0.95$) at the sensor wavelength.

- Low self-emission ($\varepsilon < 0.05$) at the sensor wavelength.
- Mechanical and thermal stability at the chamber operating temperature.

These requirements are satisfied by choosing a window material that is transparent at the sensor wavelength and applying AR coatings to minimize reflection losses. For a $\lambda_s = 3.9 \mu\text{m}$ pyrometer in an oxidation furnace ($T \approx 1000^\circ\text{C}$), sapphire (transparent to $5 \mu\text{m}$, stable to $> 1000^\circ\text{C}$) is the natural choice, with single-layer Y_2O_3 AR ($n = 1.8$, near-ideal for sapphire's $n = 1.71$).

08.2.3 Radiative Cooling: Harnessing the Atmospheric Window

Daytime radiative cooling exploits the Earth's atmospheric transparency window between 8 and $13 \mu\text{m}$, through which thermal radiation can escape to the cold sink of outer space ($\sim 3 \text{ K}$ effective temperature). The net cooling power is:

$$P_{\text{cool}} = \int_0^\infty \varepsilon(\lambda) B(\lambda, T_{\text{surface}}) d\lambda - \int_0^\infty \alpha(\lambda) B(\lambda, T_{\text{atm}}) d\lambda - \alpha_{\text{sol}} P_{\text{sol}} \quad (08.13)$$

Maximizing P_{cool} requires:

- $\varepsilon \rightarrow 1$ in the 8– $13 \mu\text{m}$ window (maximize emission into the sky).
- $\alpha_{\text{sol}} \rightarrow 0$ across the solar spectrum (reject solar heating).
- $\varepsilon \rightarrow 0$ outside the atmospheric window (reject absorbed atmospheric radiation).

As_2S_3 metasurfaces on Al substrates achieve $\varepsilon > 0.95$ in the atmospheric window with $\alpha_{\text{sol}} < 0.05$, providing net cooling of 5– 8°C below ambient under direct sunlight.

08.2.4 TPV Emitters: Spectral Concentration at the Bandgap

Thermophotovoltaic systems convert thermal radiation to electricity via photovoltaic cells. The fundamental efficiency limit (Shockley–Queisser) depends on spectral matching between the emitter and the PV cell bandgap E_g :

- Photons with $E > E_g$ ($\lambda < hc/E_g$) generate electron–hole pairs; excess energy above E_g is thermalized (loss).
- Photons with $E < E_g$ ($\lambda > hc/E_g$) pass through the cell without absorption (loss unless recycled).

A **selective emitter** with narrowband emission peaked just above E_g minimizes both loss channels, achieving spectral efficiencies of 85–90% compared to 30–40% for a blackbody emitter. W/ HfO_2 Fabry–Pérot cavities and photonic crystal silicon structures are the two most mature architectures. Chapter 10 develops TPV system design in full detail.

08.3 Cold Mirrors and Dichroic Designs

08.3.1 Decoupling Heat from Light

Many semiconductor process chambers require simultaneous optical access (for alignment, metrology, or lithography) and thermal management. A **cold mirror** reflects infrared radiation while transmitting visible light, enabling:

- Optical alignment through thermal shields in RTP chambers.
- Overlay metrology through heated process stages.
- Laser delivery through high-temperature environments.

The dichroic design principle is a multilayer stack that positions the transition edge between the visible and IR bands:

$$\rho(\lambda) = \begin{cases} \text{low} & \lambda < \lambda_{\text{cut}} \quad (\text{transmit visible}) \\ \text{high} & \lambda > \lambda_{\text{cut}} \quad (\text{reflect IR}) \end{cases} \quad (08.14)$$

Typical cutoff wavelengths: $\lambda_{\text{cut}} = 0.7 \mu\text{m}$ (visible/NIR boundary) or $\lambda_{\text{cut}} = 1.5 \mu\text{m}$ (allow NIR pyrometry through the mirror).

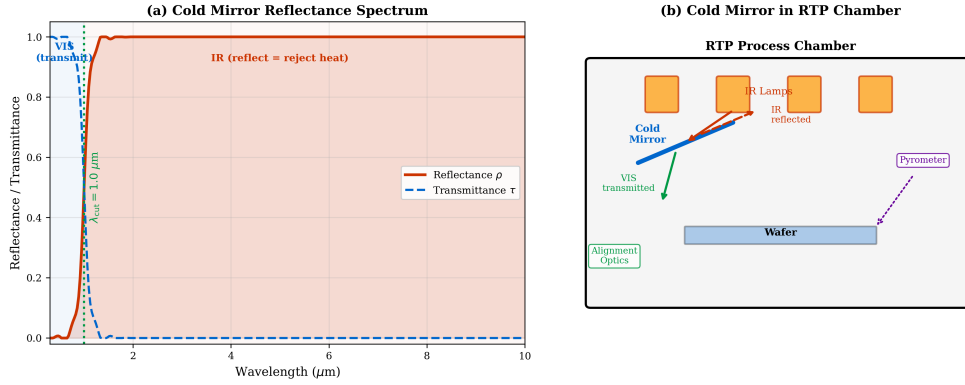


Figure 08.6: Cold mirror and dichroic filter concepts. **Left:** Reflectance spectrum of a cold mirror showing high ρ in the IR (thermal rejection) and low ρ in the visible (optical transmission). The cutoff wavelength λ_{cut} is selected based on the application. **Right:** Schematic of dichroic integration in an RTP chamber: the cold mirror allows visible alignment beams to reach the wafer while reflecting lamp IR radiation back toward the heat source, improving thermal efficiency and reducing parasitic heating of optical components.

08.3.2 Angular Dependence and Chamber Geometry

Thin-film interference coatings are inherently angle-dependent. The effective optical thickness of a layer decreases with increasing angle of incidence:

$$(nd)_{\text{eff}} = nd \cos \theta_{\text{layer}} \quad (08.15)$$

where θ_{layer} is the refraction angle inside the layer. This blue-shifts the reflectance band at oblique incidence, potentially opening transmission windows that compromise thermal performance.

Design Rule: DR-10.3 — Angular Detuning Budget

For any dichroic or cold mirror in a process chamber, specify the maximum angle of incidence and compute the resulting spectral shift from Eq. (08.15). Design the stack with sufficient bandwidth margin to maintain specification across the full angular cone. A 15° half-angle cone requires approximately 5–8% additional bandwidth beyond the normal-incidence specification.

For integrating-sphere-like chamber geometries where radiation impinges from all directions, a purely thin-film approach becomes impractical. In such cases, metallic substrates with selective absorption coatings provide broader angular acceptance at the cost of reduced spectral selectivity.

08.4 Metalenses, Diffractive Optical Elements, and Compact Spectral Control

08.4.1 Why Flat Optics for Thermal Photonics?

The spectral engineering tools developed in §08.1–§08.3 rely on thin-film multilayer stacks that modify $\varepsilon(\lambda)$ through interference in the thickness direction. These stacks are spectrally powerful

but spatially uniform: they apply the same spectral transfer function to every point on the aperture. Many production thermal problems, however, demand simultaneous spectral *and* spatial control—focusing thermal radiation from a distributed source onto a localized absorber, splitting spectral bands into different spatial channels, or shaping the angular emission profile of a selective emitter.

Classical refractive optics (lenses, prisms) can provide this spatial control, but they are incompatible with the extreme environments of semiconductor thermal processing. A 50 mm-diameter germanium plano-convex lens for LWIR focusing has a center thickness of ~ 10 mm, mass of ~ 100 g, and introduces thermal lensing that scales with the cube of the thickness (§08.6). In a process chamber operating above 600°C , the mechanical mount, differential thermal expansion, and absorbed power budget make bulk refractive elements impractical.

Metalenses and diffractive optical elements (DOEs) resolve this conflict. By encoding the phase profile in a sub-wavelength-thick structured layer ($h < 5\ \mu\text{m}$ for LWIR), they provide:

- **Spatial phase control:** Focusing, beam steering, and wavefront shaping identical to bulk optics.
- **Negligible thermal mass:** The structured layer contributes $< 0.1\%$ of the substrate thermal mass.
- **Zero thermal lensing in the structured layer:** The dn/dT -induced OPD in a $2.5\ \mu\text{m}$ nanopost layer is $\Delta W \sim 10^{-5} \times 1 \times 2.5 \times 10^{-6} = 0.025\ \text{pm}$ —eight orders of magnitude below any optical tolerance (DR-10.8, §08.4.6).
- **Spectral-spatial co-design:** The nanopost geometry can simultaneously provide phase control *and* spectral filtering.

08.4.2 Phase Coverage and Material Selection

A metalens achieves spatial phase control by varying the local phase delay $\phi(x, y)$ across the aperture. Full 2π phase coverage requires:

$$h_{\min} = \frac{\lambda_0}{\Delta n} \quad (08.16)$$

where $\Delta n = n_{\text{post}} - n_{\text{surround}}$ is the index contrast between the nanopost material and the surrounding medium (air or low-index fill).

Table 08.7: Metalens phase coverage for LWIR ($\lambda_0 = 10\ \mu\text{m}$). High-index materials achieve 2π coverage with sub- $3\ \mu\text{m}$ posts.

Material	n ($10\ \mu\text{m}$)	Δn	$h_{\min}$ (μm)	Substrate
PbTe	5.5	4.2 (vs. CaF_2)	2.38	CaF_2
Ge	4.0	2.6 (vs. BaF_2)	3.85	BaF_2
Si	3.4	2.0 (vs. SiO_2)	5.00	SiO_2
ZnSe	2.4	1.0 (vs. BaF_2)	10.0	BaF_2

PbTe on CaF_2 provides the thinnest posts and the highest fabrication margin. The 5% design margin ($h = 2.5\ \mu\text{m}$ vs. $h_{\min} = 2.38\ \mu\text{m}$) accommodates etch non-uniformity while maintaining full phase coverage.

08.4.3 AR Co-Design for Metalens Efficiency

The metalens has two optically distinct surfaces:

1. **Structured face (nanopost side):** The effective index depends on fill fraction and is spatially varying. A separate AR coating is impractical; instead, the nanopost duty cycle is co-optimized to provide partial impedance matching. Typical structured-face reflectance: $R_{\text{struct}} \approx 5\text{--}10\%$.

2. **Flat face (substrate backside):** A conventional broadband AR can be applied. Options include:
- Single-layer QW coating (e.g., ZnS on CaF₂).
 - Moth-eye sub-wavelength texture ($R < 0.5\%$).

Design Rule: DR-10.9 — Metalens AR Budget

For every metalens or DOE element in a thermal photonics system:

- Allocate and track the AR budget separately for the structured face and the flat face.
- Co-optimize the nanopost fill fraction for both phase accuracy and Fresnel loss minimization.
- Apply broadband AR (moth-eye or multilayer) to the flat face; target $R_{\text{flat}} < 1\%$.
- Report total system transmittance as $\tau_{\text{sys}} = (1 - R_{\text{struct}})(1 - R_{\text{flat}})(1 - A_{\text{bulk}}) \cdot \eta_m$, where η_m is the diffraction efficiency.

08.4.4 Tolerance Analysis for IR Metalenses

The relaxed tolerance of LWIR ($\lambda_0 = 10 \mu\text{m}$) compared to visible ($\lambda = 0.5 \mu\text{m}$) is a $20\times$ advantage in absolute terms, but the high refractive index of IR materials partially offsets this through the Δn factor.

Table 08.8: Tolerance budget for a PbTe metalens on CaF₂ at $\lambda_0 = 10 \mu\text{m}$, $f = 50 \text{ mm}$, $D = 50 \text{ mm}$.

Error Source	Sensitivity	Tolerance	WFE Contrib.
Etch depth (Δz)	$\delta\phi = (2\pi/\lambda_0)\Delta n \Delta z$	$\pm 50 \text{ nm}$	$\lambda/48$
Lateral CD (Δw)	$\delta\phi \propto (\partial n_{\text{eff}}/\partial w) \Delta w$	$\pm 20 \text{ nm}$	$\lambda/25$
Decenter (Δx)	Coma $\sim \Delta x \cdot R_{\text{ap}}^2/2f^2$	$\pm 20 \mu\text{m}$	$\lambda/4$
Tilt ($\Delta\theta$)	Astigmatism	$\pm 0.5^\circ$	$\lambda/10$
RSS total			$\lambda/3.5$

08.4.5 Alignment Mark Strategy

Unlike visible-wavelength lithography where alignment marks are inspected through the substrate at short wavelengths, IR metalens fabrication on IR-transparent substrates (Ge, ZnSe, CaF₂, chalcogenide glass) presents both a challenge and an opportunity:

- **Challenge:** The substrates are opaque in the visible. Ge is opaque below $\lambda \approx 1.8 \mu\text{m}$; standard alignment microscopes ($\lambda \approx 0.5 \mu\text{m}$) cannot see through the substrate for back-side alignment.
- **Opportunity:** At IR wavelengths ($\lambda > 2 \mu\text{m}$), the substrates are highly transparent. An IR alignment microscope operating at $\lambda = 4 \mu\text{m}$ (InSb camera) can image through the full substrate thickness, enabling direct front-to-back registration.
- **Mark design:** Cross-shaped marks ($100 \times 100 \mu\text{m}^2$, arm width $10 \mu\text{m}$) patterned in the PbTe layer during the metalens etch step serve as both intra-layer and inter-layer alignment references.
- **Placement:** Marks at ≥ 4 positions outside the clear aperture but within the substrate footprint, at 0° , 90° , 180° , and 270° on the aperture perimeter. The minimum of four marks enables extraction of translations (Δx , Δy), rotation ($\Delta\theta$), and scale (magnification error).

Design Rule: DR-10.7b — Alignment Mark Protocol for IR Metalens Fabrication

For multilayer IR metalens or DOE fabrication:

- Place ≥ 4 alignment marks at the aperture perimeter, 90° apart, in the first patterned layer.
- Use cross marks with arm width $\geq \lambda_0$ and total size $\geq 10\lambda_0$ for robust detection.
- For front-to-back registration on IR-transparent substrates, use an IR alignment microscope ($\lambda = 2\text{--}5\ \mu\text{m}$) to image through the substrate.
- Include ≥ 2 etch-depth monitor pads (unpatterned squares, $200 \times 200\ \mu\text{m}^2$) outside the clear aperture for in-situ reflectometric depth control.
- Verify total wavefront error budget: $\sigma_{\text{WFE}} = \sqrt{\sigma_{\text{decenter}}^2 + \sigma_{\text{tilt}}^2 + \sigma_{\text{depth}}^2} < \lambda_0/4$, where each σ contribution is computed from the tolerances in Table 08.8.

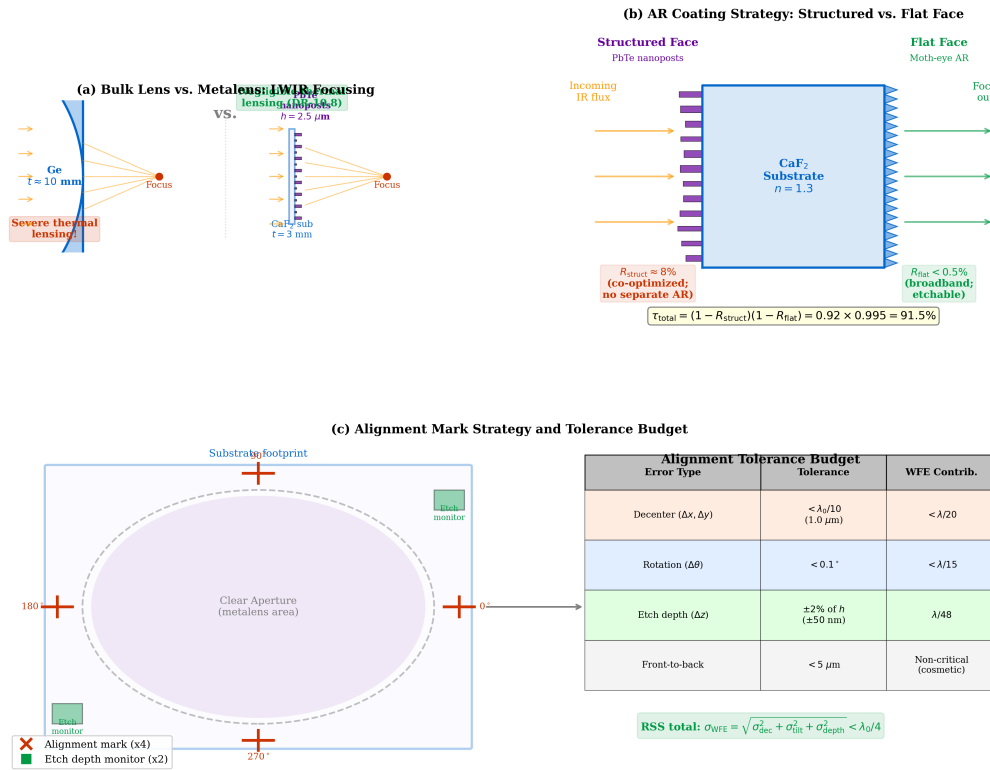


Figure 08.7: Metalens and DOE integration for thermal photonics. **(a)** Phase profile comparison: a bulk Ge lens (left, $t \approx 10\ \text{mm}$, severe thermal lensing) vs. a PbTe metalens (right, $h \approx 2.5\ \mu\text{m}$, negligible thermal lensing) achieving the same LWIR focusing function. The metalens nanopost array provides 2π phase coverage through diameter variation. **(b)** AR coating challenge: schematic cross-section showing the structured face (co-optimized nanoposts, no separate AR) and flat face (moth-eye broadband AR). Fresnel losses at each interface are annotated. **(c)** Alignment mark strategy for multilayer metalens fabrication. Four alignment marks at aperture corners enable decenter, tilt, and rotation extraction. Etch depth is controlled by in-situ reflectometry on dedicated test pads. The tolerance budget (Table 08.8) ensures that the combined wavefront error remains below $\lambda/4$ at the operating wavelength.

08.4.6 Network Context: Metalens and DOE in the MPM

A metalens or DOE element enters the Mode-Path Matrix identically to a thin window (§08.7), but with three modifications:

1. **Conduction path:** $G_{\text{cond}}^{(\text{ML})} = \kappa_{\text{sub}} A / t_{\text{sub}}$. The nanopost layer is thermally negligible ($t_{\text{post}}/t_{\text{sub}} < 0.001$).
2. **Radiation path:** The effective transmittance τ_{eff} must account for diffraction efficiency, AR losses, and the spectral weighting:

$$G_{\text{trans}}^{(\text{ML})} = 4 \tau_{\text{eff}}^{(\text{ML})} \eta_m \sigma_{\text{SB}} T^3 A F_{12} \quad (08.17)$$

where η_m is the diffraction efficiency of the target order.

3. **Thermal lensing:** Effectively zero. The dn/dT -induced OPD in a $2.5 \mu\text{m}$ nanopost layer on a 5 mm substrate is $\Delta W_{\text{ML}} = |dn/dT| \Delta T t_{\text{post}} \approx 10^{-5} \times 1 \times 2.5 \times 10^{-6} = 2.5 \times 10^{-11} \text{ m} = 0.025 \text{ pm}$ —eight orders of magnitude below any optical tolerance. The substrate thermal lens (§08.6) dominates entirely.

Design Rule: DR-10.8 — Metalens Thermal Lensing Exemption

For metalens and DOE elements with nanopost height $h < 10 \mu\text{m}$, the structured-layer contribution to thermal lensing is negligible ($\Delta W < \lambda/10^4$). Thermal lensing analysis (§08.6, DR-10.5) need only consider the substrate. This eliminates the primary advantage of bulk refractive optics (high optical power per surface) as a trade against thermal lensing: metalenses provide equivalent optical power with zero thermal lensing penalty in the structured layer.

08.5 High-Temperature Stable Coatings

08.5.1 Material Selection for Thermal Stability

Process chamber temperatures above 600°C impose severe constraints on coating materials. Oxidation, interdiffusion, delamination, and crystallization can degrade spectral performance within hours of exposure. Table 08.9 summarizes the maximum operating temperatures and transparency ranges for production-grade IR window and coating materials.

Table 08.9: IR window and coating materials for high-temperature semiconductor environments. T_{max} is the maximum continuous operating temperature for > 1000 hour lifetime.

Material	T_{max} ($^\circ\text{C}$)	Transparency	n ($10 \mu\text{m}$)	Notes
Sapphire	1000+	$0.2\text{--}5 \mu\text{m}$	1.71	Excellent mechanical; limited LWIR
ZnSe (CVD)	300	$0.5\text{--}20 \mu\text{m}$	2.40	Broad IR; moderate thermal limit
Ge	200 (coated)	$2\text{--}14 \mu\text{m}$	4.00	AR coating required; strong dn/dT
CaF ₂	600	$0.15\text{--}10 \mu\text{m}$	1.30	Hygroscopic; lowest dn/dT
GASIR	250	$1\text{--}14 \mu\text{m}$	2.50	Moldable via PGM
Si	500 (oxidation)	$1.2\text{--}8 \mu\text{m}$	3.42	Free-carrier absorption limits

Design Rule: DR-10.4 — High-Temperature Material Selection

For IR spectral elements operating above 300°C , verify that *both* the substrate and all coating layers remain below their T_{max} values (Table 08.9) with at least 50°C margin. Above 600°C , sapphire (if MWIR coverage is sufficient) or refractory oxide coatings on metallic substrates are the only production-grade options. Below 300°C , the full IR material palette (chalcogenides, II-VI semiconductors, fluorides) is available.

Fabrication Challenges

Coating degradation mechanisms at high temperature:

1. **Oxidation:** Ge coatings develop GeO_2 surface layers above 200°C in oxidizing atmospheres, reducing n from 4.0 to ~ 1.6 and destroying spectral performance. Protective Si_3N_4 or diamond-like carbon (DLC) overcoats extend Ge lifetime to 400°C .
2. **Interdiffusion:** PbTe/ZnS multilayers exhibit Pb diffusion into ZnS above 250°C , broadening the interfaces and reducing stopband rejection. Diffusion barrier layers (SiO_2 , $\sim 10\text{ nm}$) mitigate this at the cost of reduced index contrast.
3. **Crystallization:** Amorphous chalcogenide films (As_2S_3 , As_2Se_3) crystallize above their glass transition temperature ($T_g \approx 200\text{--}250^\circ\text{C}$ for As_2S_3), introducing scattering losses and spectral degradation.
4. **Delamination:** Thermal expansion mismatch between coating and substrate generates interfacial stress. For a coating of thickness d and expansion mismatch $\Delta\alpha$, the stress is approximately $\sigma \approx E_{\text{film}} \Delta\alpha \Delta T / (1 - \nu)$, where E_{film} is the film modulus and ν is Poisson's ratio.

08.6 Thermal Lensing in IR Windows

08.6.1 The dn/dT Effect

Every optical element in a process chamber absorbs some fraction of the radiative flux passing through it. This absorbed power creates a temperature gradient within the element, and through the thermo-optic coefficient dn/dT , this gradient becomes a refractive index gradient. The result is a **thermal lens**: an unintended optical power that distorts the transmitted wavefront.

For a window of thermal conductivity κ , beam radius w , absorbed power P , and thermo-optic coefficient dn/dT , the thermal lens focal length is:

$$f_{\text{th}} = \frac{\pi \kappa w^2}{P |dn/dT|} \quad (08.18)$$

This expression follows from the parabolic temperature profile created by uniform absorption in a cylindrical window with edge cooling. The temperature distribution $T(r) = T_{\text{edge}} + (P/4\pi\kappa t)(w^2 - r^2)$ produces a quadratic OPD, equivalent to a thin lens.

08.6.2 Wavefront Error Budget and Strehl Ratio

The thermal lens produces a peak-to-edge OPD of:

$$\Delta W_{\text{PV}} = \frac{w^2}{2 f_{\text{th}}} = \frac{P |dn/dT|}{2\pi \kappa} \quad (08.19)$$

For a parabolic OPD profile (power/defocus aberration), the relationship between peak-to-valley and RMS wavefront error is:

$$\Delta W_{\text{RMS}} = \frac{\Delta W_{\text{PV}}}{\sqrt{12}} \approx \frac{\Delta W_{\text{PV}}}{3.46} \quad (08.20)$$

The Strehl ratio for small wavefront errors ($\Delta W_{\text{RMS}} < \lambda/4$) is given by the Maréchal approximation:

$$S \approx \exp \left[- \left(\frac{2\pi \Delta W_{\text{RMS}}}{\lambda} \right)^2 \right] = \exp \left[- \frac{4\pi^2}{12} \left(\frac{\Delta W_{\text{PV}}}{\lambda} \right)^2 \right] \quad (08.21)$$

Trilogy Connection: Eikonal Bridge Ch. 5

The wavefront error budget methodology and Strehl ratio criterion are developed rigorously in Book 1 (*The Eikonal Bridge*, Chapter 5). The thermal lensing contribution computed here feeds directly into the system WFE budget as one component alongside surface figure errors, material inhomogeneity, and alignment errors. The $\lambda/10$ PV tolerance used throughout this chapter corresponds to $S > 0.998$ for pure defocus—well above the Maréchal criterion ($S > 0.8$ at $\lambda/4$ RMS).

08.6.3 Thermal Lensing Figure of Merit

The material-dependent factor in Eq. (08.18) defines the **thermal lensing figure of merit**:

$$F_{\text{TL}} = \frac{\kappa}{|dn/dT|} \quad (08.22)$$

Higher F_{TL} means longer thermal focal length (less thermal lensing) for the same absorbed power and geometry.

Table 08.10: Thermal lensing figure of merit for IR window materials. $F_{\text{TL}} = \kappa/|dn/dT|$.

Material	dn/dT (10^{-6} K $^{-1}$)	κ (W/m·K)	F_{TL} (10^6 W/m)	Comment
ZnSe	+61	18	0.30	Best LWIR; moderate FTL
Ge	+396	60	0.15	High n ; worst dn/dT
CaF ₂	−11.5	10	0.87	Best FTL; limited IR range
Sapphire	+13	46	3.54	MWIR only; excellent FTL
Si	+160	148	0.93	Acceptable; not broadband IR
GASIR	+72	0.4	0.006	Never use for transmissive!

Design Rule: DR-10.5 — Thermal Lensing Budget

For IR windows under > 1 W absorbed power, compute $f_{\text{th}} = \pi\kappa w^2/(P|dn/dT|)$. Require $f_{\text{th}} > 10 \times z_R$ ($10\times$ the Rayleigh range of the transmitted beam). Select materials with $F_{\text{TL}} > 0.1 \times 10^6$ W/m. Never use chalcogenide glasses in high-power transmissive paths ($F_{\text{TL}} < 0.01 \times 10^6$ W/m).

Failure Mode: Thermal Runaway in IR Windows

Thermal lensing creates a positive feedback loop when the window material has temperature-dependent absorption. If $\alpha(T)$ increases with temperature:

1. Absorbed power $P = \alpha(T) P_{\text{inc}} t$ heats the window.
2. Higher T increases α , which increases P .
3. The thermal lens strengthens, potentially focusing the beam tighter and increasing local intensity.

For Ge, α increases exponentially above 200°C due to free-carrier generation. A $100\times$ increase in α (from 5×10^{-4} to 0.05 cm^{-1}) drops f_{th} from 2.7 km to 27 m and drives the OPD from $\lambda/390$ to $\lambda/3.9$ —catastrophic failure.

08.7 Network Context: Spectral Elements in the MPM**08.7.1 How Spectral Elements Enter the Network**

Every spectral engineering element in this chapter modifies one or more conductance values in the Mode-Path Matrix. The mapping is:

- **Selective emitter/absorber:** Changes ε_{eff} on the relevant surface, modifying $G_{\text{rad}} = 4\varepsilon_{\text{eff}}\sigma_{\text{SB}}T^3A$.
- **Dichroic/cold mirror:** Splits a single radiation path into two spectral channels (reflected IR + transmitted visible), each with its own G_{rad} or G_{trans} .
- **IR window:** Creates two parallel paths—conduction through the solid and transmitted radiation through the aperture—each with distinct mode classifications.
- **Metalens/DOE:** Identical to a window but with reduced τ_{eff} (diffraction efficiency) and zero structured-layer thermal lensing.

Multiphysics Coupling

Parallel conduction–radiation paths through windows:

An IR window of thickness t , area A , conductivity κ , and transmittance $\tau(\lambda)$ creates two parallel thermal paths:

1. **Conduction path:** $G_{\text{cond}} = \kappa A/t$ (solid conduction through the window material).
2. **Transmitted radiation path:** $G_{\text{trans}} = 4\tau_{\text{eff}}(T)\sigma_{\text{SB}}T^3AF_{12}$ (radiative heat flux *through* the window from the hot interior to the cold exterior).

These are parallel paths in the thermal network. The spectral transmittance $\tau(\lambda)$ of the window (determined by its material and coatings) controls the radiation path, while κ and t control the conduction path. A good thermal window maximizes G_{trans} at diagnostic wavelengths while minimizing it at thermal wavelengths—precisely the dichroic design of §08.3.

08.7.2 Part III Verification Recipe

Every spectral engineering element in Chapters 8–10 must pass the following seven-step verification sequence before network integration. This recipe is the Part III counterpart of the conduction verification in Part II.

Reader Recipe

Part III Verification Recipe (V1–V7):

- V1. Confirm mode:** Verify $\Gamma > 10$ for the target path (Chapter 4 MPM).
- V2. Spectral specification:** Define $\varepsilon(\lambda)$ or $\tau(\lambda)$ profile from the design catalog (§08.2, Table 08.6).
- V3. Compute $\varepsilon_{\text{eff}}(T)$:** Integrate Eq. (08.4) at the operating temperature. Verify sensitivity (DR-10.6).
- V4. Material check:** Confirm all materials are below T_{max} with 50 °C margin (DR-10.4, Table 08.9).
- V5. Coating acceptance:** Verify the seven-parameter production specification (DR-10.7, Table 08.5).
- V6. Thermal lensing:** If the element is transmissive and absorbs > 1 W, compute f_{th} and verify $\Delta W < \lambda/10$ (DR-10.5, Eq. (08.18)). Compute the Strehl ratio (Eq. 08.21).
- V7. Network update:** Insert the new G_{rad} (or G_{trans}) into the MPM. Recompute Γ for all affected paths. Verify no mode reclassification has occurred.

Steady-State Limitation

Near-field limitation: The radiation conductance formula $G_{\text{rad}} = 4\varepsilon_{\text{eff}}\sigma_{\text{SB}}T^3A$ assumes far-field (Planckian) radiation exchange. When surfaces are separated by distances $d < \lambda_{\text{th}}$ (the thermal wavelength, $\lambda_{\text{th}} = b/T \approx 10 \mu\text{m}$ at 300 K), evanescent-wave coupling can enhance radiative transfer by orders of magnitude beyond the Planck limit. Near-field effects are treated in Chapter 3 (§03.5) and become relevant for sub-10 μm gaps in advanced 3D packaging. For all process chamber applications in this chapter ($d > 1 \text{ mm}$), the far-field assumption is valid.

08.8 Worked Examples

08.8.1 WE-08.1: ZnSe Window Under CO₂ Laser — Thermal Lens, WFE Budget, and MPM Integration

Problem: A ZnSe window ($\kappa = 18 \text{ W/m}\cdot\text{K}$, $n = 2.403$, $dn/dT = 6.1 \times 10^{-5} \text{ K}^{-1}$, $t = 5 \text{ mm}$, diameter $D = 50 \text{ mm}$) is used in a CO₂ laser processing chamber. The laser beam ($\lambda = 10.6 \text{ }\mu\text{m}$, $P_{\text{laser}} = 200 \text{ W}$, beam radius $w = 12 \text{ mm}$) passes through the window. The bulk absorption coefficient of the ZnSe at $10.6 \text{ }\mu\text{m}$ is $\alpha_{\text{abs}} = 5 \times 10^{-4} \text{ cm}^{-1}$.

Tasks:

- Calculate the absorbed power in the window.
- Calculate the thermal lens focal length.
- Determine whether the thermal lens exceeds the $\lambda/10$ OPD tolerance at $\lambda = 10.6 \text{ }\mu\text{m}$.
- Compute the Strehl ratio.
- Construct the MPM row for the window element.

Solution:

(a) **Absorbed power.** The absorbed fraction for a window of thickness t is:

$$f_{\text{abs}} = 1 - e^{-\alpha_{\text{abs}} t} \approx \alpha_{\text{abs}} t \quad (\text{for } \alpha_{\text{abs}} t \ll 1) \quad (\text{WE-08.1a})$$

$$\begin{aligned} f_{\text{abs}} &= 5 \times 10^{-4} \text{ cm}^{-1} \times 0.5 \text{ cm} = 2.5 \times 10^{-4} \\ P_{\text{abs}} &= 200 \text{ W} \times 2.5 \times 10^{-4} = 0.050 \text{ W} = 50 \text{ mW} \end{aligned} \quad (\text{WE-08.1b})$$

(b) **Thermal lens focal length.** From Eq. (08.18):

$$\begin{aligned} f_{\text{th}} &= \frac{\pi \times 18 \times (12 \times 10^{-3})^2}{0.050 \times 6.1 \times 10^{-5}} \\ &= \frac{\pi \times 18 \times 1.44 \times 10^{-4}}{3.05 \times 10^{-6}} \\ &= \frac{8.14 \times 10^{-3}}{3.05 \times 10^{-6}} \\ &= 2670 \text{ m} \approx 2.7 \text{ km} \end{aligned} \quad (\text{WE-08.1c})$$

(c) **OPD tolerance check.** The peak-to-valley OPD from the thermal lens is:

$$\begin{aligned} \Delta W_{\text{PV}} &= \frac{w^2}{2 f_{\text{th}}} = \frac{(12 \times 10^{-3})^2}{2 \times 2670} = \frac{1.44 \times 10^{-4}}{5340} \\ &= 2.7 \times 10^{-8} \text{ m} = 27 \text{ nm} \end{aligned} \quad (\text{WE-08.1d})$$

At $\lambda = 10.6 \text{ }\mu\text{m}$, the $\lambda/10$ tolerance is $1.06 \text{ }\mu\text{m}$. The thermal OPD of 27 nm is only $\lambda/390$ —**well within tolerance** for this clean, production-grade ZnSe.

However: if a coating defect increases local α_{abs} by $100\times$ (to 0.05 cm^{-1}), the absorbed power rises to 5 W , the focal length drops to 27 m , and the OPD increases to $2.7 \text{ }\mu\text{m} = \lambda/3.9$, *exceeding tolerance*. This is the thermal runaway scenario described in the failure box of §08.6.

(d) **Strehl ratio.** Converting PV to RMS using Eq. (08.20):

$$\Delta W_{\text{RMS}} = \frac{27 \text{ nm}}{\sqrt{12}} = \frac{27}{3.46} = 7.8 \text{ nm} \quad (\text{WE-08.1d2})$$

The Strehl ratio from Eq. (08.21):

$$\begin{aligned} S &= \exp \left[- \left(\frac{2\pi \times 7.8 \times 10^{-9}}{10.6 \times 10^{-6}} \right)^2 \right] = \exp [-(4.63 \times 10^{-3})^2] \\ &= \exp(-2.14 \times 10^{-5}) = 0.99998 \end{aligned} \quad (\text{WE-08.1d3})$$

Essentially diffraction-limited. The thermal lens contribution to the WFE budget is negligible for clean ZnSe at this power level.

For the defect scenario ($P = 5$ W): $\Delta W_{\text{RMS}} = 780$ nm, $S = \exp[-(2\pi \times 780 \times 10^{-9}/10.6 \times 10^{-6})^2] = \exp(-0.214) = 0.81$ —just above the Maréchal limit ($S = 0.8$), confirming marginal failure.

(e) MPM row for the window.

The window temperature is approximately $T \approx 400$ K (estimated from edge cooling). The window area $A = \pi(25 \times 10^{-3})^2 = 1.96 \times 10^{-3}$ m².

$$G_{\text{cond}}^{(\text{window})} = \frac{18 \times 1.96 \times 10^{-3}}{5 \times 10^{-3}} = 7.06 \text{ W/K} \quad (\text{WE-08.1e})$$

$$\begin{aligned} G_{\text{rad}}^{(\text{window})} &= 4 \times 0.3 \times 5.67 \times 10^{-8} \times 400^3 \times 1.96 \times 10^{-3} \\ &= 4 \times 0.3 \times 5.67 \times 10^{-8} \times 6.4 \times 10^7 \times 1.96 \times 10^{-3} \\ &= 8.55 \times 10^{-3} \text{ W/K} \end{aligned} \quad (\text{WE-08.1f})$$

$$\Gamma_{\text{window}} = \frac{8.55 \times 10^{-3}}{7.06} = 1.2 \times 10^{-3} \quad (\text{WE-08.1g})$$

Table 08.11: MPM row for WE-08.1 ZnSe window element.

Path	G_{cond} (W/K)	G_{rad} (W/K)	Γ	Mode	Tool	Action
Window (solid)	7.06	8.6×10^{-3}	1.2×10^{-3}	Cond.	Part II	Optimize κ , t
Window (trans.)	—	$0.23^\dagger$	—	Rad.	Part III	Spectral filter design

[†]Transmitted radiation conductance $G_{\text{trans}} = 4 \tau_{\text{eff}} \sigma_{\text{SB}} T^3 A F_{12}$ estimated with $\tau_{\text{eff}} \approx 0.7$ at 400 K.

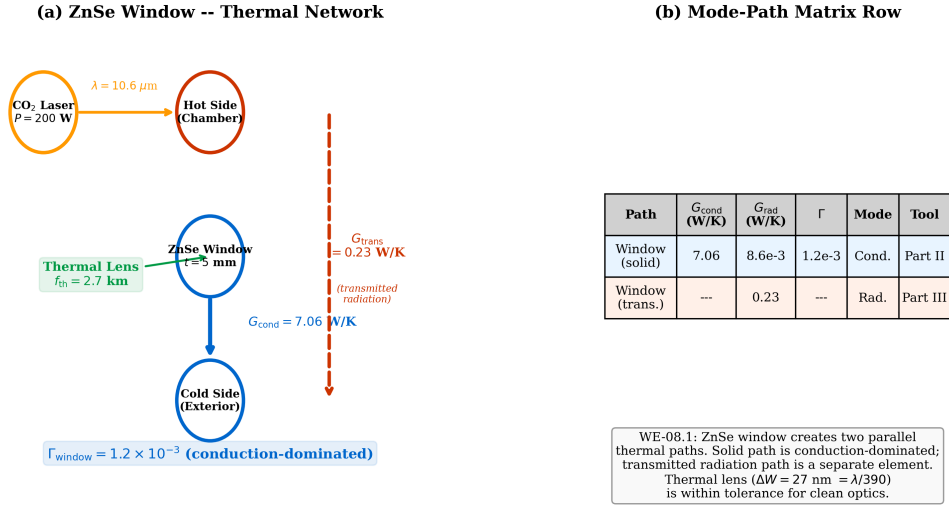


Figure 08.8: WE-08.1 network diagram. The ZnSe window creates two parallel thermal paths: a conduction path through the solid window ($G_{\text{cond}} = 7.06 \text{ W/K}$, blue) and a transmitted radiation path ($G_{\text{trans}} = 0.23 \text{ W/K}$, red). The thermal lens (green annotation) affects the optical path but not the thermal network directly. The MPM classifies the solid path as conduction-dominated ($\Gamma = 1.2 \times 10^{-3}$) and the transmitted radiation path as a separate radiation element. Strehl ratio: $S = 0.99998$ (clean optics); $S = 0.81$ (defect scenario).

08.8.2 WE-08.2: PbTe Selective Emitter for Thermophotovoltaic Application

Problem: Design a selective emitter for a GaSb TPV cell ($E_g = 0.73 \text{ eV}$, $\lambda_g = hc/E_g = 1.70 \mu\text{m}$). The emitter operates at $T = 1200 \text{ K}$ and must achieve spectral efficiency $\eta_{\text{spec}} > 80\%$, defined as:

$$\eta_{\text{spec}} = \frac{\int_0^{\lambda_g} \varepsilon(\lambda) B(\lambda, T) d\lambda}{\int_0^{\infty} \varepsilon(\lambda) B(\lambda, T) d\lambda} \quad (\text{WE-08.2a})$$

Design: PbTe nanopillar array on tungsten substrate.

- **Target:** $\varepsilon > 0.9$ for $1.0 < \lambda < 1.7 \mu\text{m}$; $\varepsilon < 0.1$ for $\lambda > 2.5 \mu\text{m}$.
- **Architecture:** Subwavelength PbTe nanopillars ($h = 0.8 \mu\text{m}$, $d = 0.4 \mu\text{m}$, pitch $p = 0.6 \mu\text{m}$) on a polished W substrate.
- **Mechanism:** The PbTe nanopillars act as resonant absorbers near λ_g ; the W substrate provides low ε at longer wavelengths.

Performance estimate:

At $T = 1200 \text{ K}$, $\lambda_{\text{max}} = 2898/1200 = 2.42 \mu\text{m}$. The Planck function is significant from approximately 1 to $10 \mu\text{m}$.

For a blackbody emitter, the fraction of power below $\lambda_g = 1.70 \mu\text{m}$ is:

$$f_{\text{BB}}(\lambda_g T) = f(1.70 \times 1200) = f(2040 \mu\text{m} \cdot \text{K}) \approx 0.07 \quad (\text{WE-08.2b})$$

using the blackbody fractional function. A blackbody spectral efficiency is only $\sim 7\%$ —disastrously low.

With the selective emitter ($\varepsilon = 0.9$ below λ_g , $\varepsilon = 0.05$ above $2.5 \mu\text{m}$):

$$\eta_{\text{spec}} \approx \frac{0.9 \times 0.07}{0.9 \times 0.07 + 0.05 \times 0.93} = \frac{0.063}{0.063 + 0.0465} = \frac{0.063}{0.1095} \approx 0.575 \quad (\text{WE-08.2c})$$

This falls short of the 80% target with a simple two-level model. Achieving $> 80\%$ requires either:

- A back-surface reflector ($\rho > 0.95$) on the TPV cell to recycle sub-bandgap photons (Chapter 10).
- Reducing the long-wavelength ε below 0.02 (requires more aggressive cavity Q or photonic crystal design).

MPM integration: The selective emitter modifies G_{rad} for the emitter→cell path:

$$\begin{aligned} G_{\text{rad}}^{(\text{selective})} &= 4 \times \varepsilon_{\text{eff}}(1200 \text{ K}) \times \sigma_{\text{SB}} \times 1200^3 \times A \\ \varepsilon_{\text{eff}}(1200 \text{ K}) &\approx 0.9 \times 0.07 + 0.05 \times 0.93 = 0.110 \end{aligned} \quad (\text{WE-08.2d})$$

Compare to the blackbody case: $\varepsilon_{\text{eff}} = 1.0$, so the selective emitter *reduces* total G_{rad} by a factor of ~ 9 —but concentrates the useful (above-bandgap) fraction by a factor of $0.9/0.07 \times 0.07/0.110 \approx 5.7$.

08.8.3 WE-08.3: RTP Chamber — Before/After Spectral Coating MPM Update

Problem: An RTP chamber operates at $T = 1100 \text{ K}$ with three thermal paths:

- **Path A:** Lamp → wafer (radiation, $\varepsilon_{\text{eff}} = 0.7$, $A = 0.07 \text{ m}^2$, $F_{12} = 0.6$).
- **Path B:** Wafer → chamber wall (radiation, $\varepsilon_{\text{eff}} = 0.5$, $A = 0.07 \text{ m}^2$, $F_{12} = 0.3$).
- **Path C:** Wafer → chuck (conduction through gas gap, $G_{\text{cond}} = 5 \text{ W/K}$).

A low-emissivity coating ($\varepsilon_{\text{eff}} = 0.1$) is applied to the chamber walls to reduce parasitic radiation loss. Compute the full MPM before and after, and assess the impact on wafer temperature uniformity.

Solution — Step-by-step MPM analysis:

Step 1: Compute G_{rad} for each radiation path (before coating).

$$\begin{aligned} G_{\text{rad}}^{(\text{A})} &= 4 \times 0.7 \times 5.67 \times 10^{-8} \times 1100^3 \times 0.07 \times 0.6 \\ &= 4 \times 0.7 \times 5.67 \times 10^{-8} \times 1.331 \times 10^9 \times 0.042 \\ &= 4 \times 0.7 \times 5.67 \times 10^{-8} \times 5.59 \times 10^7 \\ &= 4 \times 0.7 \times 3.170 = 8.88 \text{ W/K} \end{aligned} \quad (\text{WE-08.3a})$$

$$\begin{aligned} G_{\text{rad}}^{(\text{B,before})} &= 4 \times 0.5 \times 5.67 \times 10^{-8} \times 1100^3 \times 0.07 \times 0.3 \\ &= 4 \times 0.5 \times 5.67 \times 10^{-8} \times 1.331 \times 10^9 \times 0.021 \\ &= 4 \times 0.5 \times 5.67 \times 10^{-8} \times 2.795 \times 10^7 \\ &= 4 \times 0.5 \times 1.585 = 3.17 \text{ W/K} \end{aligned} \quad (\text{WE-08.3b})$$

Step 2: Mode classification (before coating).

Using Path B radiation as the radiation contribution for the wafer–chuck path:

$$\Gamma_{\text{C}} = \frac{G_{\text{rad}}^{(\text{B,before})}}{G_{\text{cond}}^{(\text{C})}} = \frac{3.17}{5.0} = 0.63 \quad (\text{WE-08.3c})$$

This places Path C in the *mixed mode* regime ($0.1 < \Gamma < 10$). Both conduction and radiation tools are needed.

Step 3: Apply wall coating ($\varepsilon_{\text{eff}} : 0.5 \rightarrow 0.1$).

$$\begin{aligned} G_{\text{rad}}^{(\text{B,after})} &= 4 \times 0.1 \times 5.67 \times 10^{-8} \times 1100^3 \times 0.07 \times 0.3 \\ &= 4 \times 0.1 \times 1.585 = 0.634 \text{ W/K} \end{aligned} \quad (\text{WE-08.3d})$$

Step 4: Mode reclassification (after coating).

$$\Gamma_{C,\text{after}} = \frac{0.634}{5.0} = 0.127 \quad (\text{WE-08.3e})$$

Still mixed mode, but now barely above the 0.1 threshold. The radiation contribution has been suppressed by a factor of 5 $\times$.

Step 5: Uniformity impact.

The parasitic radiation loss through Path B acts as a spatially non-uniform heat sink (chamber wall temperature varies). The non-uniformity contribution scales with the $G_{\text{rad}}^{(B)}$ magnitude:

$$\delta T_{\text{non-uniform}} \propto \frac{G_{\text{rad}}^{(B)} \cdot \delta T_{\text{wall}}}{G_{\text{rad}}^{(A)} + G_{\text{cond}}^{(C)}} \quad (\text{WE-08.3f})$$

Assuming $\delta T_{\text{wall}} = 50$ K variation:

$$\delta T_{\text{before}} \approx \frac{3.17 \times 50}{8.88 + 5.0} = \frac{158.5}{13.88} = 11.4 \text{ K} \quad (\text{WE-08.3g})$$

$$\delta T_{\text{after}} \approx \frac{0.634 \times 50}{8.88 + 5.0} = \frac{31.7}{13.88} = 2.3 \text{ K} \quad (\text{WE-08.3h})$$

The wall coating reduced temperature non-uniformity from ± 11.4 K to ± 2.3 K—a 5 $\times$ improvement, directly traceable to the 5 $\times$ reduction in $G_{\text{rad}}^{(B)}$.

Step 6: Updated MPM summary.

Table 08.12: WE-08.3: MPM before and after wall coating.

Path	Before G_{rad} (W/K)	After G_{rad} (W/K)	Γ (before)	Γ (after)	Mode
A: Lamp $\rightarrow$ wafer	8.88	8.88	—	—	Rad.
B: Wafer $\rightarrow$ wall	3.17	0.634	—	—	Rad.
C: Wafer $\rightarrow$ chuck	$G_{\text{cond}} = 5.0$		0.63	0.127	Mixed $\rightarrow$ Mixed

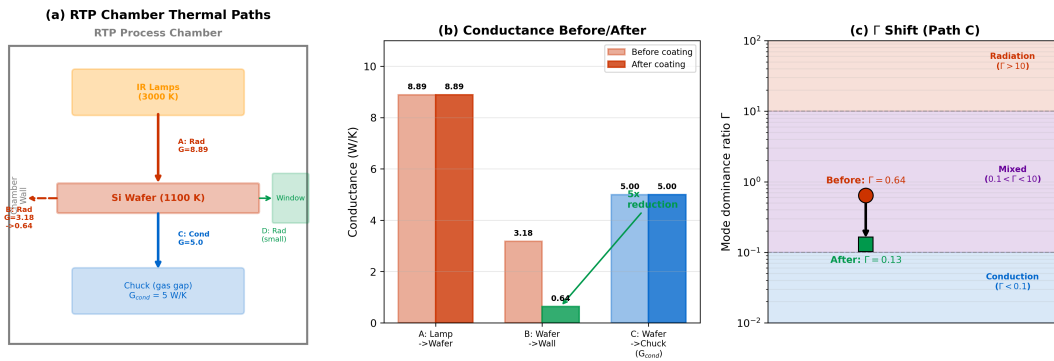


Figure 08.9: WE-08.3: RTP chamber MPM before and after wall coating application. **(a)** Before: parasitic radiation path B ($G_{\text{rad}} = 3.17$ W/K) creates mixed-mode coupling ($\Gamma = 0.63$) and ± 11.4 K non-uniformity. **(b)** After: low- ε wall coating reduces path B to 0.634 W/K ($\Gamma = 0.127$), suppressing non-uniformity to ± 2.3 K. **(c)** Γ shift diagram showing the before/after mode classification for Path C.

Key takeaway: A single spectral modification (wall coating) changed the system behavior from mixed-mode with significant radiation interference to nearly conduction-dominated, reducing temperature non-uniformity by 5 $\times$. The MPM before/after comparison makes the engineering benefit quantitatively explicit.

08.8.4 WE-08.4: PbTe Metalens for LWIR Thermal Concentrator — AR Budget and Alignment Tolerance

Problem: A PbTe metalens on a CaF₂ substrate is designed to focus LWIR radiation ($\lambda_0 = 10 \mu\text{m}$) from a 50 mm-diameter collection aperture to a 5 mm-diameter thermopile detector, achieving 10× geometric concentration. The metalens has focal length $f = 50 \text{ mm}$, nanopost height $h = 2.5 \mu\text{m}$ on a $t = 3 \text{ mm}$ CaF₂ substrate. The back face has a moth-eye AR texture.

Tasks:

- Verify 2π phase coverage.
- Compute the AR transmission budget.
- Determine the decenter tolerance for $< \lambda/4$ wavefront error.
- Compute the etch depth tolerance for $< \lambda/10$ phase error.
- Construct the MPM row.

Solution:

(a) Phase coverage. Maximum index contrast: $\Delta n = n_{\text{PbTe}} - n_{\text{CaF}_2} = 5.5 - 1.3 = 4.2$. Minimum height for 2π : $h_{\min} = \lambda_0/\Delta n = 10/4.2 = 2.38 \mu\text{m}$. Design height $h = 2.5 \mu\text{m}$ provides 5% margin. ✓

(b) AR transmission budget.

- Structured face (PbTe nanoposts): co-optimized fill fraction gives $R_{\text{struct}} \approx 8\%$ averaged over 8–14 μm .
 - Flat face (CaF₂ moth-eye): $R_{\text{flat}} \approx 0.5\%$.
 - Bulk absorption in CaF₂ ($\alpha \approx 10^{-3} \text{ cm}^{-1}$ at 10 μm , $t = 3 \text{ mm}$): $A_{\text{bulk}} = 1 - e^{-10^{-3} \times 0.3} \approx 0.03\%$.
- $$\tau_{\text{total}} = (1 - 0.08)(1 - 0.005)(1 - 0.0003) = 0.92 \times 0.995 \times 1.00 = 0.915 \quad (\text{WE-08.4a})$$

This exceeds the $\tau > 0.90$ requirement (DR-10.9c). ✓

(c) Decenter tolerance. For a metalens of focal length f and aperture radius R_{ap} , a lateral decenter Δx introduces coma-like wavefront error:

$$\Delta W_{\text{coma}} \approx \frac{\Delta x \cdot R_{\text{ap}}^2}{2f^2} \quad (\text{WE-08.4b})$$

For $\Delta W_{\text{coma}} < \lambda_0/4 = 2.5 \mu\text{m}$:

$$\Delta x < \frac{2.5 \times 10^{-6} \times 2 \times (50 \times 10^{-3})^2}{(25 \times 10^{-3})^2} = \frac{2.5 \times 10^{-6} \times 5.0 \times 10^{-3}}{6.25 \times 10^{-4}} = 20 \mu\text{m} \quad (\text{WE-08.4c})$$

(d) Etch depth tolerance. Phase error from depth error:

$$\delta\phi = \frac{2\pi}{\lambda_0} \Delta n \Delta z = \frac{2\pi}{10} \times 4.2 \times \Delta z \quad (\text{WE-08.4d})$$

For $\delta\phi < 2\pi/10$ (i.e., $\lambda/10$ wavefront error): $\Delta z < 10/(10 \times 4.2) = 0.238 \mu\text{m} = 238 \text{ nm}$. At $\pm 2\%$ of $h = 2.5 \mu\text{m}$: $\Delta z = \pm 50 \text{ nm}$, giving $\delta\phi = 2\pi \times 4.2 \times 0.05/10 = 0.132 \text{ rad} = \lambda/48$. Well within tolerance. ✓

(e) MPM row.

Table 08.13: MPM row for WE-08.4 PbTe metalens element.

Path	G_{cond} (W/K)	G_{rad} (W/K)	Γ	Mode
ML substrate (cond)	$6.54^\dagger$	5.3×10^{-3}	8.1×10^{-4}	Cond.
ML aperture (trans. rad)	—	$0.18^\ddagger$	—	Rad.

$^\dagger \text{CaF}_2$: $\kappa = 10 \text{ W/mK}$, $A = \pi(25 \text{ mm})^2 = 1.96 \times 10^{-3} \text{ m}^2$, $t = 3 \text{ mm}$.

$^\ddagger \tau_{\text{eff}} = 0.915$, $\eta_m = 0.85$ (est.), $T = 400 \text{ K}$, $F_{12} = 0.8$.

Key insight: The metalens substrate dominates the conduction path (identical to a plain window). The metalens *nanopost layer* contributes zero measurable thermal lensing (DR-10.8) but reduces τ_{eff} by $\sim 8\%$ through structured-face Fresnel losses— the single largest efficiency penalty, motivating the AR co-design emphasis of DR-10.9.

Design Rules Summary

Table 08.14: Complete set of spectral engineering and thermal window design rules (v2: 10 rules).

Rule	Section	Description
DR-10.1	§08.1	For radiation-dominated paths ($\Gamma > 10$), the design variable is $\varepsilon_{\text{eff}}(T)$, computed from Planck-weighted spectral emissivity. Network element: $G_{\text{rad}} = 4\varepsilon_{\text{eff}}\sigma_{\text{SB}}T^3A$.
DR-10.2	§08.2.1	Match spectral emissivity profile to application using the design catalog (Table 08.6). Verify $\varepsilon_{\text{eff}}(T)$ within linearization range.
DR-10.3	§08.3	For dichroic/cold mirror designs, include angular detuning budget. A 15° half-angle cone requires 5–8% additional spectral bandwidth.
DR-10.4	§08.5	For IR elements above 300°C , verify all materials below T_{max} with 50°C margin. Above 600°C , only sapphire or refractory oxides are production-grade.
DR-10.5	§08.6	For IR windows under $> 1\text{ W}$ absorbed power, compute $f_{\text{th}} = \pi\kappa w^2/(P dn/dT)$. Require $f_{\text{th}} > 10 \times z_R$. Select $F_{\text{TL}} > 0.1 \times 10^6\text{ W/m}$. Never use chalcogenides in high-power transmissive paths.
DR-10.6	§08.1.3	Sensitivity recomputation threshold: recompute ε_{eff} and update G_{rad} in the MPM whenever $\Delta T > 50\text{ K}$ or $ \Delta\varepsilon_{\text{eff}} > 0.05$. Either triggers $> 10\%$ conductance change.
DR-10.7	§08.1.5	Coating production acceptance: every spectral coating must pass the seven-parameter specification (Table 08.5) including thermal cycling ($ \Delta\varepsilon_{\text{eff}} < 0.02$ after 10^3 cycles), film stress, and emissivity drift.
DR-10.7b	§08.4.5	Alignment mark protocol for multilayer IR metalens/DOE fabrication: ≥ 4 perimeter marks, IR through-substrate alignment, depth monitor pads, and a closed WFE budget target $\sigma_{\text{WFE}} < \lambda_0/4$.
DR-10.8	§08.4.6	Metalens thermal lensing exemption: for nanopost height $h < 10\text{ }\mu\text{m}$, structured-layer $\Delta W < \lambda/10^4$. Thermal lensing analysis need only consider the substrate.
DR-10.9	§08.4.3	Metalens AR budget: allocate AR losses separately for the structured face and flat face; target $R_{\text{flat}} < 1\%$, co-optimize duty cycle to reduce R_{struct} , and report τ_{sys} including bulk absorption and diffraction efficiency.

Looking Forward

This chapter established the spectral engineering toolkit for radiation-dominated paths ($\Gamma > 10$): emissivity as the central design variable, thin-film interference for spectral shaping, dichroic designs for heat–light decoupling, metalenses for spatial–spectral co-control, material selection for high-temperature stability, thermal lensing mitigation for IR windows, and a quantitative sensitivity framework (DR-10.6) for determining when spectral elements require re-evaluation.

Every spectral element enters the Mode-Path Matrix as a modified radiation conductance $G_{\text{rad}} = 4\varepsilon_{\text{eff}}\sigma_{\text{SB}}T^3A$.

The Part III verification recipe (V1–V7) standardizes the integration workflow for all spectral elements across Chapters 8–10.

Chapter 9 applies these tools to two critical applications: radiative cooling (where the atmospheric transparency window provides the heat sink) and multiphysics cloaking (where both conduction and radiation paths must be blocked simultaneously). Chapter 9 also addresses the mixed-mode regime ($0.1 < \Gamma < 10$), where neither the Part II nor Part III toolkit alone suffices and coupled analysis is required. WE-08.3 provided a preview of this mixed-mode coupling: the RTP chamber wall coating shifted the system from $\Gamma = 0.63$ to $\Gamma = 0.127$, demonstrating how spectral engineering can push a mixed-mode system toward conduction dominance.

Chapter 10 extends the spectral engineering framework to thermophotovoltaics, where the selective emitter principles developed here combine with back-surface reflectors and near-field enhancements to achieve high-efficiency thermal-to-electrical energy conversion.

For sub-10 μm gaps in advanced 3D packaging, the far-field radiation conductance formula used throughout this chapter must be replaced by the near-field formulation of Chapter 3 (§03.5). The spectral engineering principles—emissivity control, material selection, coating design—remain applicable, but the transfer function changes from Planckian to evanescent-wave dominated.

Research Roadmap

Table 08.15: Research roadmap for Chapter 8 topics.

Topic	Current State	5-Year Horizon	TPD Connection
Nonreciprocal thermal emission ($\varepsilon \neq \alpha$)	Lab demos with magneto-optic materials	Integrated radiative management devices	Breaks Kirchhoff symmetry; new DR needed
High-T metasurfaces	Refractory metal (W, Mo) nanopillars	Oxide-protected selective emitters to $> 1500^\circ\text{C}$	Extends Table 08.9 range
Inverse design of spectral profiles	Genetic algorithm + TMM	Topology optimization of $\varepsilon(\lambda)$ stacks	Automated MPM-to-coating pipeline
Thermal GRIN compensation	Ad hoc refocus	Co-design tools linking FEA to ray trace	Eikonal Bridge Ch. 5 integration
Dynamic emissivity surfaces	VO_2 phase-change ($T_c \approx 68^\circ\text{C}$)	Electrically switchable ε arrays	Merges Ch. 6 switchable + Ch. 8 spectral
In-situ ε_{eff} monitoring	Post-process FTIR measurement	Real-time spectral feedback during deposition	Closes the loop on DR-10.7 acceptance

Problems

- 8.1. Effective emissivity calculation.** A tungsten filament ($\varepsilon \approx 0.03$ at $\lambda = 1 \mu\text{m}$; $\varepsilon \approx 0.3$ at $\lambda = 10 \mu\text{m}$, with approximately linear spectral increase) operates at $T = 2800 \text{ K}$. (a) Calculate the Wien peak wavelength. (b) Estimate ε_{eff} by dividing the spectrum into SWIR, MWIR, and LWIR bands and weighting by the Planck fractional function. (c) How does ε_{eff} change at $T = 1200 \text{ K}$?
- 8.2. Quarter-wave stack design.** Design a $(\text{H L})^N$ stack to achieve $\rho > 0.95$ over $8\text{--}12 \mu\text{m}$ using Ge/ZnS. (a) Compute λ_0 (center wavelength) and $\Delta\lambda/\lambda_0$ from Eq. (08.11). (b) Determine the minimum N (number of periods) needed for $\rho > 0.95$ at λ_0 . (c) Estimate the total stack thickness. (d) What happens if ZnSe ($n = 2.4$) is substituted for ZnS ($n = 2.2$) as the low-index material?
- 8.3. Cold mirror angular performance.** A cold mirror with $\lambda_{\text{cut}} = 1.5 \mu\text{m}$ is designed using a Si/SiO₂ stack ($n_H = 3.4$, $n_L = 1.4$). (a) Compute the spectral shift at $\theta = 30^\circ$ incidence for both TE and TM polarizations. (b) Determine the required bandwidth increase to maintain specification for a $\pm 15^\circ$ cone. (c) Sketch the reflectance curve at 0 , 15 , and 30° .
- 8.4. Thermal lensing comparison.** A pyrometry window ($D = 30 \text{ mm}$, $t = 3 \text{ mm}$) absorbs $P = 2 \text{ W}$ from stray lamp radiation (beam fills the full aperture). Compare the thermal lens focal length and OPD for: (a) ZnSe, (b) Ge, (c) CaF₂, and (d) sapphire using Table 08.10. Which material is optimal for a $\lambda = 3.9 \mu\text{m}$ pyrometer?
- 8.5. Selective emitter optimization.** For a GaSb TPV cell ($E_g = 0.73 \text{ eV}$) at $T = 1200 \text{ K}$: (a) Calculate η_{spec} for a two-level emitter with $\varepsilon_{\text{high}} = 0.95$ below λ_g and $\varepsilon_{\text{low}} = 0.02$ above $2.5 \mu\text{m}$. (b) Compute the ε_{eff} and G_{rad} per unit area. (c) Compare to a blackbody emitter. (d) How does a back-surface reflector with $\rho = 0.95$ for $\lambda > \lambda_g$ change the effective spectral efficiency? (Preview of Chapter 10.)
- 8.6. MPM update with spectral coating.** An RTP chamber has three thermal paths at $T = 1100 \text{ K}$:
- Path A: Lamp→wafer (radiation, $\varepsilon_{\text{eff}} = 0.7$, $A = 0.07 \text{ m}^2$, $F = 0.6$).
 - Path B: Wafer→chamber wall (radiation, $\varepsilon_{\text{eff}} = 0.5$, $A = 0.07 \text{ m}^2$, $F = 0.3$).
 - Path C: Wafer→chuck (conduction through gas gap, $G_{\text{cond}} = 5 \text{ W/K}$).
- (a) Compute G_{rad} for paths A and B. (b) Compute Γ for path C (using path B's G_{rad} as the radiation contribution). (c) Which path is the critical path (highest thermal resistance)? (d) A selective coating is applied to the chamber wall to reduce ε_{eff} from 0.5 to 0.1 . Recompute G_{rad} for path B and determine the new critical path.
- 8.7. Material selection for a pyrometry window.** A pyrometer operates at $\lambda = 3.9 \mu\text{m}$ in an oxidation furnace at $T = 1000^\circ\text{C}$. The window must have:
- $\tau > 0.90$ at $3.9 \mu\text{m}$.
 - Stability above 1000°C .
 - $F_{\text{TL}} > 0.5 \times 10^6 \text{ W/m}$.
- (a) Using Tables 08.9 and 08.10, identify candidate materials. (b) For each candidate, check the transparency range at $3.9 \mu\text{m}$. (c) Which material(s) satisfy all three requirements? (d) If no single material qualifies, propose a composite design (substrate + coating).

8.8. Metalens AR and alignment budget. A Ge metalens ($n = 4.0$, nanopost height $h = 2.0 \mu\text{m}$, $\lambda_0 = 10 \mu\text{m}$) on a ZnSe substrate ($n = 2.4$, $t = 5 \text{ mm}$) focuses LWIR radiation with $f = 30 \text{ mm}$ and aperture diameter $D = 25 \text{ mm}$. (a) Calculate the minimum nanopost height for 2π phase coverage and verify the design margin. (b) Compute Fresnel reflection losses at each interface (structured face, flat face) assuming no AR coating. (c) If a single-layer ZnS AR coating ($n = 2.2$) is applied to the flat face at $\lambda_0 = 10 \mu\text{m}$, calculate the QW thickness and the resulting reflectance. (d) Determine the lateral decenter tolerance for $\lambda/4$ wavefront error. (e) Determine the etch depth tolerance for $\lambda/10$ phase error. (f) If the moth-eye AR on the flat face reduces R_{flat} to 0.3%, compute the total system transmittance assuming $R_{\text{struct}} = 10\%$. (g) Construct the MPM row for this element at $T = 500 \text{ K}$.

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Chapter 8 of “Thermal Photonics: A Unified Framework for Engineering Heat Flux”

Book 3 of the Open-Source Engineering Optics Trilogy

Cross-references: Book 1 The Eikonal Bridge Ch. 3, 5, 8 (thin-film design, WFE budget, coating fabrication); Book 2 Quantum Field Imaging Ch. 4 (detector radiometry).